

DANIEL SCHUURMAN

Department of Agriculture and Resource Economics
University of California, Davis

Email: dschuurman@ucdavis.edu
Phone: (916)-701-7181

Education

Ph.D in Agriculture and Resource Economics, University of California, Davis

Expected June 2027

M.Sc in Food, Agriculture and Resource Economics, University of Guelph

2021

Fields of Interest

Agricultural Economics, Climate Economics, and Econometrics

References

Dalia Ghanem (**Co-chair**) - Associate Professor
Dept. of Agriculture and Resource Economics
University of California, Davis
dghanem@ucdavis.edu

Rachael Goodhue (**Co-chair**) - Professor
Dept. of Agriculture and Resource Economics
University of California, Davis
regoodhue@ucdavis.edu

Aaron Smith - Professor
Dept. of Agriculture and Resource Economics
University of California, Berkeley
aaronsmith@berkeley.edu

Research in Progress

Yesterday Matters: Accounting for Feedback in Acreage Responses to Climate Change (Job Market Paper)

Abstract: Understanding the potential impacts of climate change on agricultural systems requires knowing how quickly producers reallocate land in response. When adjustments are costly, adaptation to climate change may unfold gradually. Existing research commonly uses static models that omit information on past acreage decisions. Using a dynamic panel framework that accounts for feedback from lagged allocations, I identify considerable persistence in crop acreage allocations. This persistence has a first-order effect on how climate-change projections should be constructed. Due to the persistence, future acreage allocations depend on the cumulative effects along the entire transition path of climate, a fact disregarded by projections conducted with static models. Acreage forecasts under CMIP6 scenarios demonstrate that predictions with static models overestimate acreage responses by an average of 19-22% under moderate and high emissions scenarios, relative to the dynamic model, by the year 2050. Consequently, traditional static methods overestimate the pace of adaptation in settings with persistent adjustment frictions.

To What Extent Can Long-differencing Capture Climate Adaptation? **Draft**
with Dalia Ghanem and Felix Pretis

Abstract: Understanding the degree to which we are able to adapt to climate change is central to economic assessments of future climate damages. Economists increasingly use comparisons between long differences and fixed effects estimators to measure climate adaptation. We show that such comparisons can be misleading. Neither estimator is consistent for its intended parameter, as both the long-difference (LD) and fixed effects (FE) estimands are weighted averages of the long- and short-run responses to climate and weather. As a result, the difference between the two understates the true extent of adaptation, and the standard test based on this difference –while controlling size – tends to be substantially underpowered in the settings researchers typically encounter. An empirically-calibrated simulation shows this difference understates adaptation by 20–70%, depending on the averaging window.

Abstract: We estimate the causal effects of pesticide exposure on prenatal health in California. We assemble an individual-level dataset from 2007 to 2017 linking birth outcomes to fine-scaled daily local pesticide exposure. Using quasi-experimental variation in exposure via pesticide drift determined by location-specific wind direction on the day-of-application, we find downwind exposure leads to adverse impacts on birth weight and gestation duration. The detected effects from pesticide drift are consistent with pesticides that exhibit high volatilization.

Publications

Heterogeneity, Climate Change and Crop Yield Distributions: Solvency Implications for Publicly-Subsidized Crop Insurance Programs. Published in the American Journal of Agricultural Economics in 2025 with Alan Ker.
Winner of 2025 Best Paper Award in the Applied Risk Analysis Section of the AAEA.

Business Risk Management Programs and the Adoption of Beneficial Management Practices in Canada. Published in the Canadian Journal of Agricultural Economics in 2024 with Aaron De Laporte, Tristan Skolrud, Peter Slade, and Alfons Weersink.

Research Experience

Graduate Student Researcher <i>with Dalia Ghanem</i>	Jan. 2025 - Aug. 2026
Graduate Student Researcher <i>with Rachael Goodhue</i>	Sept. 2022 - Dec. 2024
Researcher Assistant <i>with Alan Ker</i>	June 2021 - Aug. 2022

Teaching Experience

Graduate Teaching Assistant: <i>Econometrics for Business Decisions (4.8/5.0)</i>	Winter Quarter 2026
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Scholarships, Honors, and Awards

2026	UC Davis Summer Research Fellowship
2025	UC Davis Jastro-Shields Graduate Research Award
2024	UC Davis Summer Research Fellowship; UC Davis Jastro-Shields Graduate Research Award
2021	<i>(Pre-PhD)</i> Ontario Graduate Scholarship, Graduate Tuition Scholarship
2020	<i>(Pre-PhD)</i> Board of Graduate Studies Research Scholarship, Farm Managers and Rural Appraisers Award, Dr. Zuhair Ahmad Hassan Memorial Graduate Scholarship, Soden Memorial Graduate Scholarship in Agriculture

Professional Activities

Conference and Seminar Presentations

* indicates upcoming presentations

2026	SCC-76 Annual Meeting, World Congress of Environment and Resource Economics (WCERE), *American Applied Economic Association Annual Meeting (AAEA)
2025	UC Davis Agricultural Economics Seminar
2022	<i>(Pre-PhD)</i> USDA ERS Brownbag Seminar
2021	<i>(Pre-PhD)</i> AAEA; CAES; Climate Smart Soils Conference

Service

2026	Chamberlain Seminar in Econometrics, Student Organizer
2025	UC Davis Agricultural Economics Seminar Series, Student Organizer; ARE Faculty Search Committee, Student Representative

Technical Reports

Costs and Benefits of On-Farm Beneficial Management Practices that Reduce Net GHG Emissions.

Prepared for Farmers for Climate Solutions report, *Rooted in Climate Action: Recommendations for the next Agricultural Policy Framework*, with Aaron De Laporte and Alfons Weersink (June 2022).

Business Risk Management Programs and the Adoption of Higher Resilience and Lower Emissions Practices in Canada. Prepared for Farmers for Climate Solutions report, *Grounded in Resilience: Adapting Business Risk Management Programs to Reward Climate-Friendly Agriculture*, with De Laporte A., Peter Slade, Tristan Skolrud, Alfons Weersink, Katherine Lorimer, and Ryan Tougas-Cooke (February 2022).

Towards a Business Case for Soil Health: A Synthesis of Current Knowledge on the Economics of Soil Health Practices in Ontario. Prepared for The Greenbelt Foundation with Aaron De Laporte, Alfons Weersink, Claudia Wagner-Riddle and Paul Smith (February 2022).

Costs and Benefits of Effective and Implementable On-Farm Beneficial Management Practices that Reduce Greenhouse Gases. Prepared for Farmers for Climate Solutions, *A Down Payment for a Resilient Farm Future: Budget 2021 Recommendation*, with Aaron De Laporte and Alfons Weersink (February 2021).